DELHI PUBLIC SCHOOL JANKIPURAM, LUCKNOW

SESSION 2025-2026 : PERIODIC ASSESSMENT-III

Subject : Mathematics (241)

SET–II

Time : 3 Hrs.

M.M. : 80

Date :_____ Name : _____ Class/Sec X_____

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General Instructions:

Read the following instructions carefully and follow them:

All questions are compulsory

Section A: Question 1-18 are objective type and conceptual based and question 19 and 20 are assertion and reason type each consist of 1 mark each.

Section B: Question 21-25 consist of 2 marks each.

Section C: Question 26-31 consist of 3 marks each.

Section D: Question 32-35 consist of 5 marks each.

Section E: Question 36-38 consist of 4 marks each containing two case study based questions with subparts of 1+1+2 marks.

Internal choices are provided in each section. Use of calculator is not permitted.

This question paper consists of 6 pages.

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Section - A

Section A consists of 20 questions of 1 mark each

1. What is the greatest possible speed at which a man can walk 52 km and 91 km in an exact number of hours?

   (a) 17 km/hours (b) 7 km/hours (c) 13 km/hours (d) 26 km/hours

2. The pair of linear equations $2x + 5y = 7$, $6x - 5y = 11$ has a unique solution, if represents two lines which are

   (a) intersecting at exactly one point. (b) parallel.

   (c) coincident. (d) intersecting at exactly two points.

3. The empirical relationship between the three measures of central tendency is :

   (a) 3 Median = Mode + 2 Mean (c) 3 Median = Mode – 2 Mean

   (b) 3 Mode = Median + 2 Mean (d) 3 Mean = Mode – 2 Median

4. If the quadratic equation $x^2 + 4x + k = 0$ has real and equal roots, then

   (a) $k < 4$ (b) $k > 4$ (c) $k = 4$ (d) $k \geq 4$

5. If the radii of two circles are in the ratio of **4 : 3**, then their areas are in the ratio of :

   (a) 4 : 3 (b) 8 : 3 (c) 16 : 9 (d) 9 : 16

6. Which of the following is an incorrect statement ?

   (a) Two congruent triangles are also similar.

   (b) A square and a rhombus are not similar.

   (c) Two triangles are similar if their corresponding sides are proportional.

   (d) Two polygons are similar if and only if their corresponding sides are in proportion.

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7. A tree casts a shadow 15 m long on the level of ground, when the angle of elevation of the sun is 45^0. The height of a tree is____.

(a) 10 m (b) 14 m

(c) 8 m (d) 15 m

8. Consider the following frequency distribution

Class	0 – 5	5 – 10	10 – 15	15 – 20	20 – 25
Frequency	11	12	13	9	11

The upper limit of median class is_____

(a) 10 (b) 13 (c) 15 (d) 20

9. If the first term of an A.P. is 'a' and its common difference is 'b', then its 10th term is :

(a) a + 10b (b) 10a + b

(c) a + 9b (d) 9a + b

10. Which of the following statements is not true?

a) A number of secants can be drawn at any point on the circle.

b) Only one tangent can be drawn at any point on a circle.

c) A chord is a line segment joining two points on the circle

d) From a point inside a circle only two tangents can be drawn.

11. If ABC is right angled at C, then the value of cos(A + B) is____

(a) 0 (b) 1 (c) 90^0 (d) 45^0

12. Ratio of lateral surface areas of two cylinders with equal height is

(a) 1 : 2 (b) H : h

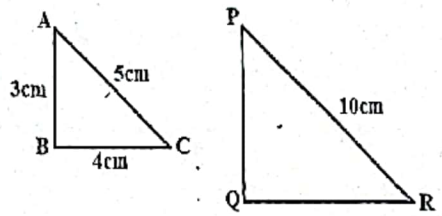
(c) R : r (d) None of these

13. Find: $2\sin 30°\sin 90° - 3\cos^2 60°\sec^2 30°$

(a) 1 (b) None of these

(c) (a) and (b) (d) 3

14. If $\Delta ABC \sim \Delta PQR$, then perimeter of the triangle PQR (in cm) is



(a)12 (b)24

(c) 18 (d)20

15. The centroid of the triangle whose vertices are **(3, - 7),(- 8, 6) and (5, 10)** is___.

(a) (0, 9 (b) (0, 3)

(c) (1, 3) (d) (3, 5)

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16. The probability of the happening of an event is 'p' and the probability of non-happening of the same event is 'q'. The relation between 'p' and 'q' is :

(a) $p+q+1=0$ (b) $p=q-1$

(c) $p+q=1$ (d) $p=1, q=1$

17. The co-ordinates of the point which is mirror image of the point $(-3, 5)$ about x-axis are

(a) $(3, 5)$ (c) $(3, -5)$

(b) $(-3, -5)$ (d) $(-3, 5)$

18. Which one of the following is not equal to Unity?

(a) $\sin^2 x + \cos^2 x$ (b) $\cot^2 x - \operatorname{cosec}^2 x$

(c) $\sec^2 x - \tan^2 x$ (d) $\tan x . \cot x$

Direction for questions 19 & 20: In question numbers 19 and 20, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

(a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(b) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

(c) Assertion (A) is true but Reason (R) is false.

(d) Assertion (A) is false but Reason (R) is true.

19. Assertion (A) : If two triangles are equiangular, then they are similar.

Reason (R) : If two triangles are similar, then they are congruent

20. Assertion: The value of sinA is always less than and equal to 1

Reason: sin A is the product of sin and A

SECTION - B

Section B consists of 5 questions of 2 marks each.

21. The length of the minute-hand of a clock is 14 cm. Find the area swept by this minute-hand in 5 minutes.

OR

To warn ships for underwater rocks, a lighthouse throws a red colored light over a sector of central angle 80° up to a distance of 16·5 km. Find the area of the sea over which the ships are warned.

22. **If $\sqrt{3}\tan\theta = 1$, then find the value of $\sin^2\theta - \cos^2\theta$**

23. Find the value of p if the pair of equations $2x+3y-5=0$ and $px-6y-8=0$ has a unique solution.

24. If the zeroes of the quadratic polynomial $x^2+(a+1)x+b$ are 2 and -3, then find the values of a and b.

OR

A cottage industry produces a certain number of pottery articles in a day. It was observed on a particular day that the cost of production of each article (in rupees) was 3 more than twice the number of articles produced on that day. If the total cost of production on that day was Rs 90, find the number of articles produced and the cost of each article.

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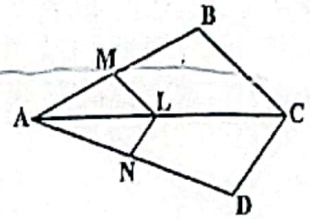
25. Prove that $\frac{AM}{MB} = \frac{AN}{ND}$ if LM || CB and LN || CD as shown in the figure.



SECTION - C

Section C consists of 6 questions of 3 marks each.

26. If the point Q(0, 1) is equidistant from points P(5, – 3) and R(x, 6), then find the value/s of x. Also, find the distance QR and PR.

27. In a class test, the sum of marks obtained by Ananya in Mathematics and Science is 28. Had she got 3 marks more in Mathematics and 4 marks less in Science, the product of marks obtained in the two subjects would have been 180. Find the marks obtained in the two subjects separately

28. a)From a pack of 52 playing cards, jack, queen and king of diamonds are removed. A card is drawn at random from the remaining cards. Find the probability of getting a face card or a card of spades.

b) Two dice are thrown at the same time. Find the probability of getting different number on both dice.

29. Prove that $7 + 2\sqrt{5}$ is irrational.

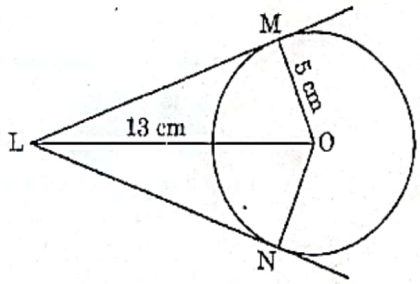
30. Prove the following identities, where the angles involved are acute angles for which the expressions are defined

$$\frac{\cos A - \sin A + 1}{\cos A + \sin A - 1} = \operatorname{cosec} A + \cot A,$$

OR

Prove that : $(\sin x - \cos x + 1).(\sec x - \tan x) = (\sin x + \cos x)$

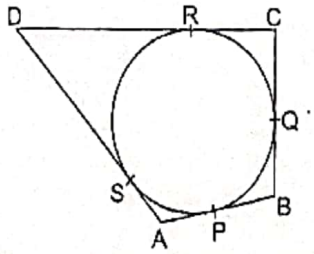
31. In the given figure, from a point L which is at a distance of 13 cm from the centre O of a circle of radius 5 cm, the pair of tangents LM and LN are drawn to the circle. Find the perimeter of quadrilateral LMON.



OR

A quadrilateral ABCD is drawn to circumscribe a circle, as shown in the given : Show that

$$\frac{AB + CD}{AD + BC} = 1$$



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Section D

Section D consists of 4 questions of 5 marks each.

32. Check graphically whether the pair of linear equations $2x + 3y = 6$; $3x + 4y = 12$ is consistent. If so, solve it graphically. Find the solution of the equations from the graph.

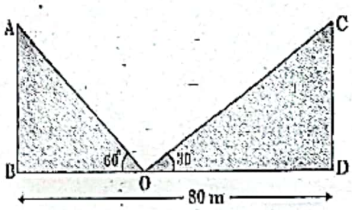
OR

A train travels at a certain average speed for a distance of 132 km and then travels a distance of 140 km at an average speed of 4 km/h more than the initial speed. If it takes 4 hours to complete the whole journey, what was the initial average speed? Determine the time taken by train to cover the distances separately.

33. A TV tower stands vertically on a bank of a canal. From a point on the other bank directly opposite the tower, the angle of elevation of the top of the tower is 60°. From a point 20 m away from this point on the same bank, the angle of elevation of the top of the tower is 30°. Find the height of the tower and the width of the canal.

OR

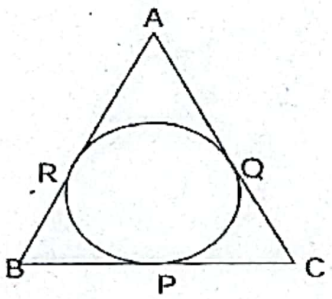
Two poles (AB and CD) of equal heights are standing opposite each other on either side of the road, which is 80 m wide. From a point O between them on the road, the angles of elevation of the top of the poles are 60° and 30°, respectively.



Answer the following questions, using the information given above.

(i) Find the distances of the point (O) from the poles.
(ii) Find the height of each pole.
(iii) Find the ratio of height of pole to the distance of point from the closer pole.

34. **Prove that the lengths of tangents drawn from an external point to a circle are equal.**
Using above result, find the length BC of ΔABC. Given that, a circle is inscribed in Δ ABC touching the sides AB, BC and CA at R, P and Q respectively and AB= 10 cm, AQ= 7cm ,CQ= 5cm.



35. The mean of the following distribution is 48 and sum of all the frequency is 50. Find the missing frequencies x and y.

Class	20-30	30-40	40-50	50-60	60-70
Frequency	8	6	x	11	y

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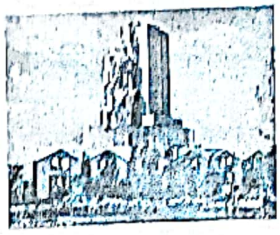
Section - E

Case study-based questions are compulsory.

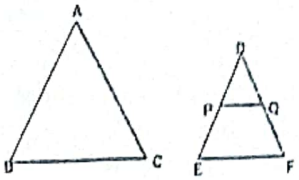


36. Triangle is a very popular shape used in interior designing. The picture given above shows a cabinet designed by a famous interior designer. Here the largest triangle is represented by $\triangle$ ABC and smallest one with shelf is represented by $\triangle$ DEF. PQ is parallel to EF.

(i) Show that $\triangle DPQ \sim \triangle DEF$.

(ii) If DP = 50 cm and PE = 70 cm then find PQ/EF.

(iii) If 2AB = 5DE and $\triangle ABC \sim \triangle DEF$ then show that $\frac{\text{Perimeter of } \triangle ABC}{\text{Perimeter of } \triangle DEF}$ is constant.



OR

If AM and DN are medians of triangles ABC and DEF respectively then prove that $\triangle ABM \sim \triangle DEN$.

37. Avil was very happy to receive a fancy jumbo pencil from his best friend Roha on his birthday. Pencil is a basic writing tool, when sharpened its shape is a combination of cylinder & cone as given in the picture. Cylindrical pencil with conical head is a common shape worldwide since ages. Commonly pencils are made up of wood & plastic but we should promote pencils made up of eco-friendly material (many options available in the market these days) to save environment.

The dimensions of Rinku's pencil are given as follows: Length of cylindrical portion is 21cm. Diameter of the base is 1 cm and height of the conical portion is 1.2 cm Based on the above information, answer the following questions:

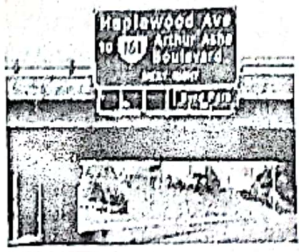
(i) Find the slant height of the sharpened part.

ii) Find curved surface area of sharpened part (in terms of π).

iii) Find the total surface area of the pencil (in terms of π).

OR

The pencil's total height decreases by 8.2 cm after sharpening it many times, what is the volume of the cylindrical part of the shortened pencil (in terms of π)?



38. In a charming village, there is a street called 'Maplewood Avenue', where the houses are numbered from 1 to 49. A community group has decided to undertake a green renovation project, aiming to install solar panels on the houses.

They have a few questions before they begin the project

(i) How many houses are there on 'Maplewood Avenue' ?

(ii) The house numbers follow an Arithmetic Progression. What is the first term of the A.P. and the common difference ?

(iii) If the group wanted to calculate the sum of all house numbers from 1 to 49, how much would that be ?

OR

What is the sum of the house numbers between 15 and 30 ?

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